

# WhatsApp Receipt Business Event Gateway Report

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Selected skill: 004-aqstoqflow-business-event-gateway

## Summary

This pass moved WhatsApp POS receipt delivery from the checkout request path into the existing business event and outbox backbone. A completed POS sale can now queue a WhatsApp receipt as a durable business event/outbox message, while a worker later claims, sends, retries, defers, or dead-letters the request.

The POS cashier experience remains unchanged: WhatsApp still requires a destination and explicit customer consent before charging. The sale is not blocked by provider availability.

## Implemented

- Added `queueWhatsAppReceiptDeliveryInTx` to record a WhatsApp receipt business event and outbox row atomically.
- Reused existing `BusinessEvent` and `BusinessEventOutbox` models without adding a migration.
- Used existing `WEBHOOK` outbox channel with event name `pos.receipt.whatsapp.requested`.
- Kept raw phone numbers out of business event payloads, metadata, audit logs, and delivery results.
- Stored the raw phone only as the outbox delivery destination required for later provider dispatch.
- Added a WhatsApp receipt worker service with tenant-scoped claim, stale-lock recovery, deferred configuration state, retry/backoff, dead-letter handling, and worker completion events.
- Added a bounded worker CLI wrapper in `scripts/whatsapp-receipt-worker.ts`.
- Updated `sendReceipt` so WhatsApp queues delivery; print, email, and SMS still use the existing provider dispatch.

## Files changed

- `services/communication/whatsapp-receipt-outbox.service.ts`
- `services/communication/whatsapp-receipt-worker.service.ts`
- `services/communication/__tests__/whatsapp-receipt-outbox.service.test.ts`
- `services/communication/__tests__/whatsapp-receipt-worker.service.test.ts`
- `scripts/whatsapp-receipt-worker.ts`
- `scripts/__tests__/whatsapp-receipt-worker.test.ts`
- `services/pos/receipt.service.ts`
- `services/pos/__tests__/receipt-public.test.ts`

## Verification

- `npm test -- --runInBand`  
`services/communication/__tests__/whatsapp-receipt-outbox.service.test.ts`  
`services/communication/__tests__/whatsapp-receipt-worker.service.test.ts`  
`scripts/__tests__/whatsapp-receipt-worker.test.ts: passed, 3 suites, 7 tests.`
- `npm test -- --runInBand`  
`services/communication/__tests__/whatsapp-receipt.provider.test.ts`  
`services/communication/__tests__/whatsapp-receipt-outbox.service.test.ts`  
`services/communication/__tests__/whatsapp-receipt-worker.service.test.ts`  
`services/pos/__tests__/receipt-channel-contract.test.ts`  
`services/pos/__tests__/receipt-public.test.ts`  
`services/pos/__tests__/pos.service.test.ts`  
`components/pos/__tests__/ProfessionalPOSSystem.shift-close.test.tsx`  
`scripts/__tests__/whatsapp-receipt-worker.test.ts: passed, 8 suites, 49 tests.`
- `npm test -- --runInBand services/pos/__tests__ actions/pos/__tests__`  
`services/events/__tests__/business-event.service.test.ts: passed, 18 suites passed, 1 skipped;`  
`138 tests passed, 7 skipped.`
- `npm run typecheck: first run timed out; rerun passed.`
- `Focused ESLint over touched WhatsApp/POS/worker files: passed.`
- `npm run prisma:validate: passed.`
- `Runtime safety scan for obvious WhatsApp test secrets/raw test phone numbers outside tests: no findings.`

## Blocked gate

`npm run policy:gates` was attempted and stopped at `inventory:boundary:fail`.

Exact failing command:

```
node scripts/inventory-boundary-gate.js --mode fail
```

Cause: the inventory boundary gate is scanning generated Next build output under `.next-dev/` and `.next-slice410/`, producing 66 direct stock mutation findings from generated server files. This blocker is unrelated to the WhatsApp receipt gateway slice.

## Notes

- Global module entitlement behavior was not changed.
- WhatsApp live sends remain opt-in via environment configuration.
- Provider dispatch is now worker-controlled, so later WhatsApp features can reuse the same queue/worker pattern rather than coupling provider calls to user-facing actions.

Next recommended numbered skill: `005-aqstoqflow-accounting-control-center`.